60.4

AIME 2024 avg@32

VAPO: Value-Model-Based RL for Advanced Reasoning

Surpassing DAPO by 10+ Points on AIME 2024

Base Model: Qwen2.5-32B · No SFT Data · ByteDance Seed

arXiv: 2504.05118

2026-04-20

Value Models Enable Precise Token-Level Credit Assignment for Long CoT

Value-free RL assigns uniform advantage across all tokens; value-model RL resolves credit at each step — critical for long reasoning chains.

Value-Free RL (GRPO / DAPO)

Trajectory reward spread uniformly across tokens

t_1 | t_2 | t_3 | t_4 | ... | t_n | R=1

Advantage assigned uniformly:

+A | +A | +A | +A | +A | +A

Limitations:

✗ Cannot identify which step caused failure

✗ High variance from Monte Carlo estimates

✗ Poor sample efficiency on complex tasks

DAPO: 50 pts on AIME 2024

State-of-the-art value-free method

Value-Based RL (VAPO)

Per-token advantage via value model V(s)

t_1 | t_2 | t_3 | t_4 | ... | t_n | R=1

Precise per-token credit:

$+A_1$ | $+A_2$ | $-A_3$ | ← error step $+A_4$ | $+A_5$

Three Key Advantages:

Credit Assignment
Token-level precision catches step-level errors in CoT

Variance Reduction
Lower-variance estimates vs. Monte Carlo per token

Generalization
Well-trained value model improves sample efficiency

VAPO: 60 pts on AIME 2024

+10 pts over DAPO

CHALLENGES

Three Unsolved Problems Block Value-Model-Based RL for Long-CoT Reasoning

Long trajectories introduce bias, heterogeneous lengths demand adaptive tradeoffs, and sparse rewards degrade credit propagation.

01 Value Model Bias

From Long Trajectories

- Long trajectories amplify bootstrap errors step-over-step
- Bias accumulates as CoT chains grow longer
- Inaccurate advantage estimates mislead policy updates

Vanilla PPO result: 5/60

Naïve value model barely works

02 Heterogeneous Lengths

Conflicting Bias-Variance Needs

- Short responses need low variance (low λ in GAE)
- Long responses need low bias (high λ in GAE)
- Fixed λ is sub-optimal across the full length range

No prior work addresses this

VAPO introduces Length-Adaptive GAE

03 Reward Signal Sparsity

Amplified by Long CoT

- Only final <eos> token receives reward signal
- Thousands of steps before any gradient signal
- Credit propagation degrades over very long sequences

Exacerbated at 6,000+ tokens

VAPO generates avg ~6K-token responses

FRAMEWORK

VAPO Systematically Addresses All Three Challenges With Seven Integrated Technique

Each VAPO component targets a specific structural weakness — eliminating the failure modes of vanilla value-model RL.

01 Value Model Bias
Long trajectory bootstrap errors

Vanilla PPO: 5 pts
VAPO w/o pretrain: 11 pts
VAPO full: 60 pts

02 Heterogeneous Lengths
Conflicting bias-variance needs

w/o Length-Adaptive GAE: 45
λ fixed → suboptimal tradeoff
Novel: λ adapts to length

03 Reward Sparsity
Only <eos> receives reward

Clip-Higher + Token Loss
Positive LM Loss + Group-Sample
Dense learning signals

Value-Pretraining
Pre-trains value model before RL to reduce initial bias

Decoupled-GAE
Robust value estimation (from VC-PPO)

Length-Adaptive GAE NOVEL
Dynamic λ per response length; short→low λ, long→high λ
−15 pts if removed

Clip-Higher
Asymmetric clipping for exploration (DAPO)

Token-Level Loss
Normalize at token level (DAPO)

Positive LM Loss
Self-imitation on correct trajectories

Group-Sampling
Multi-response per prompt (GRPO)

VAPO Ablation Results
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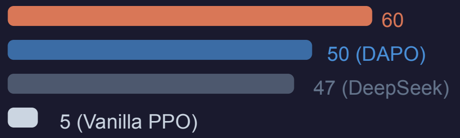
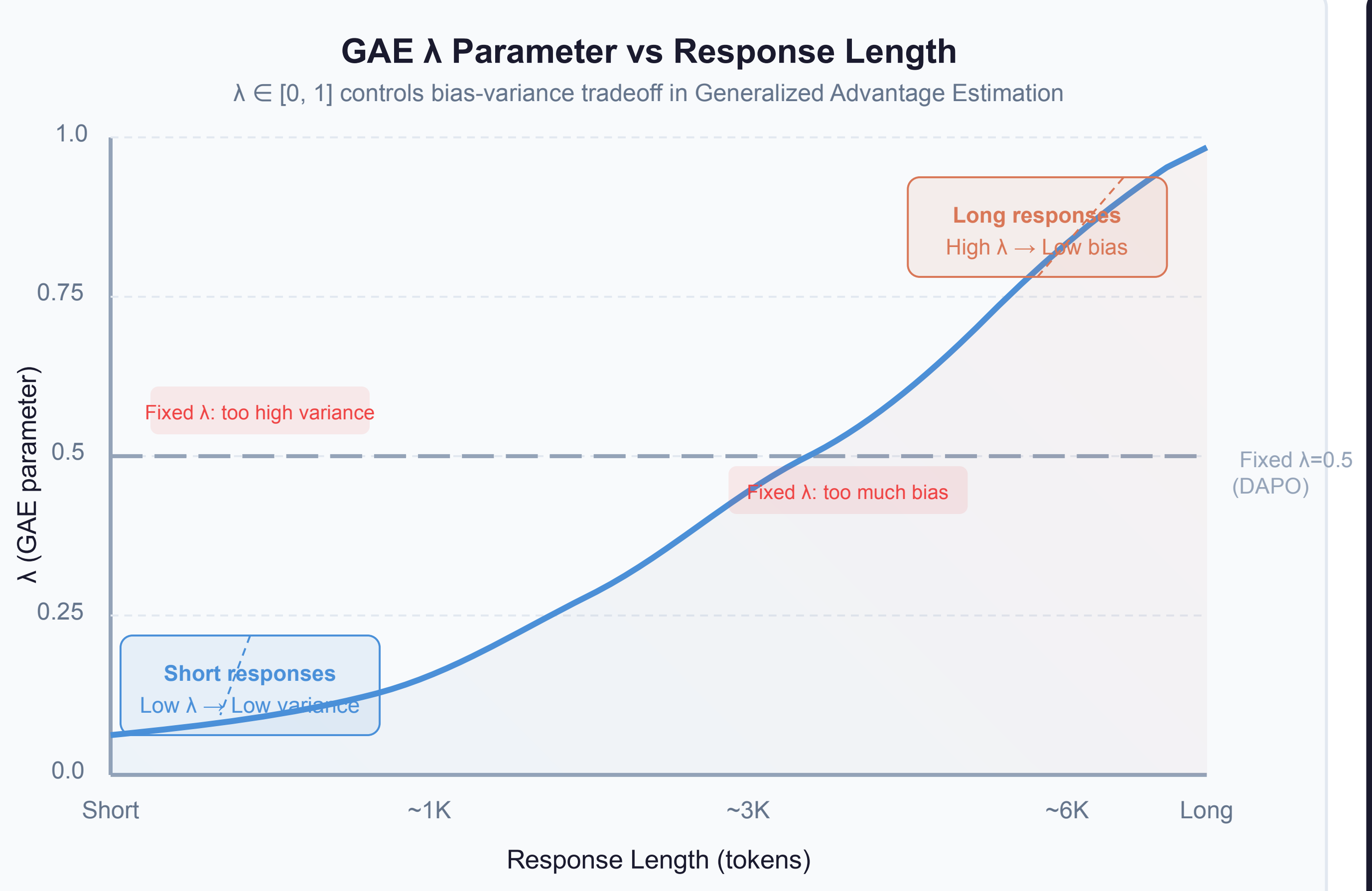
Vanilla PPO	5
w/o Value-Pretraining	11
w/o Decoupled-GAE	33
w/o Length-Adaptive GAE	45
w/o Clip-Higher	46
w/o Token-level Loss	53
w/o Positive LM Loss	54
w/o Group-Sampling	55
VAPO (full)	**60**



NOVEL TECHNIQUE

Length-Adaptive GAE: Dynamic λ Solves the Bias-Variance Tradeoff for Variable Res

Short responses → lower λ (prioritize low variance). Long responses → higher λ (prioritize low bias). One adaptive rule handles all lengths.



How Length-Adaptive GAE Works

The Problem With Fixed λ

GAE with fixed λ can't optimize for both short and long sequences simultaneously.

Prior methods (DAPO, GRPO) use fixed λ

VAPO's Solution

λ increases monotonically with response length — each sequence gets optimal λ.

One adaptive rule handles heterogeneous lengths

−15 pts if removed

VAPO: 60 → w/o Length-Adaptive GAE: 45

3rd most impactful component in ablation

Not present in DAPO, GRPO, or VC-PPO

Comparison

Method	λ type
GRPO, DAPO	Fixed λ
VC-PPO	Fixed λ
VAPO	**Adaptive λ(length)**

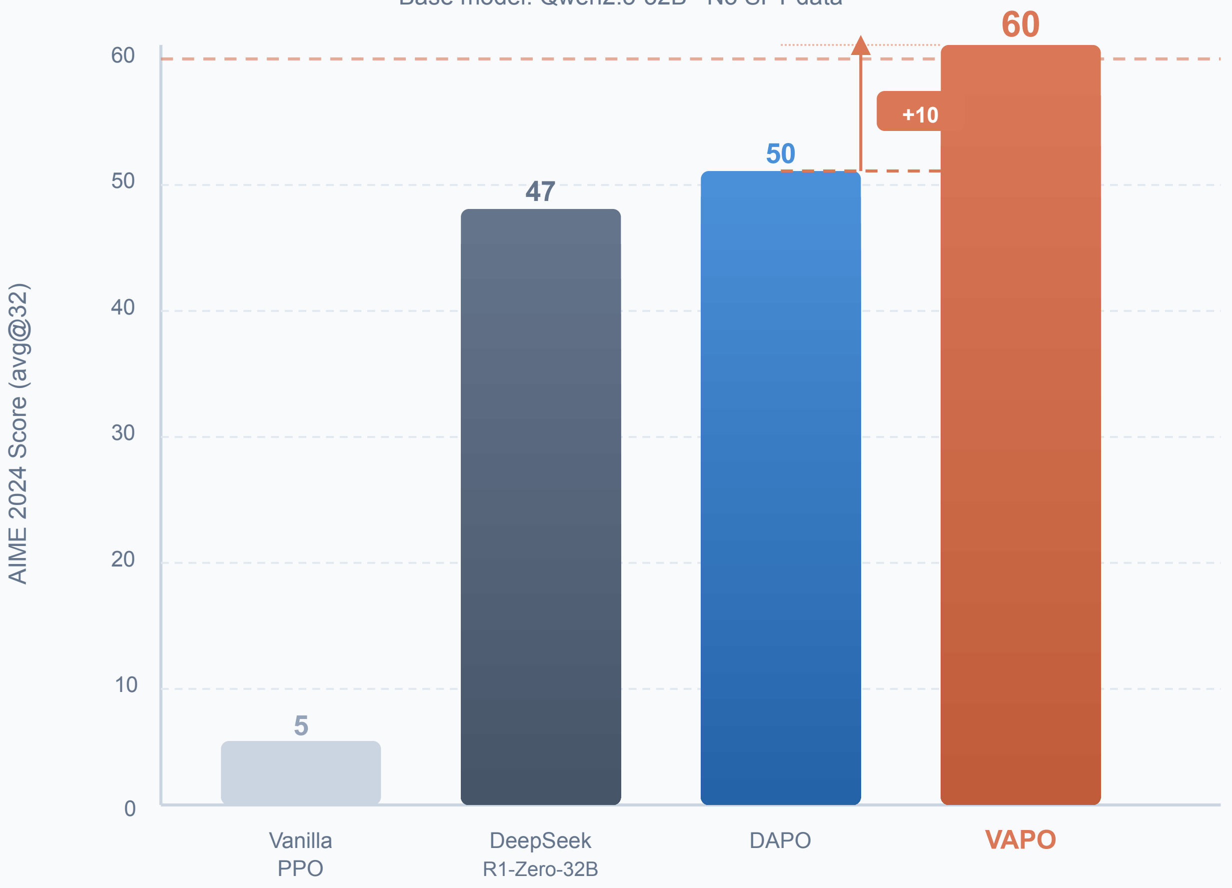
RESULTS

VAPO Scores 60 on AIME 2024 — 10+ Points Above Prior State-of-the-Art

All models trained on the same Qwen2.5-32B base, no SFT data. VAPO surpasses DAPO by +10 and DeepSeek-R1-Zero by +13 points.

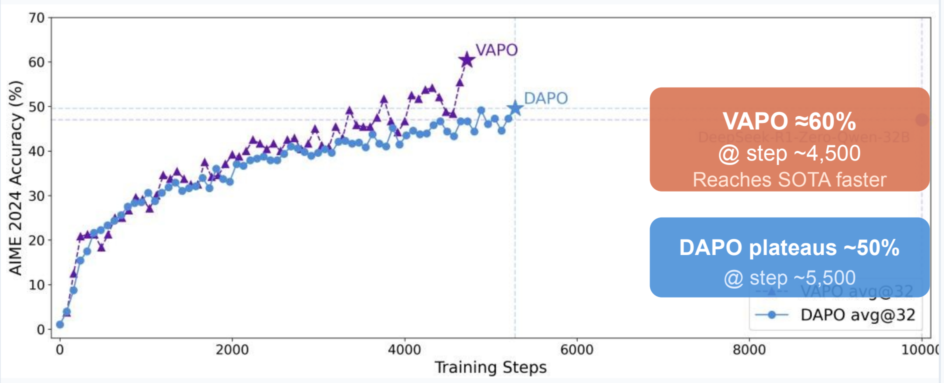
AIME 2024 avg@32 Score Comparison

Base model: Qwen2.5-32B · No SFT data



Training Accuracy vs Steps

VAPO vs DAPO vs DeepSeek-R1-Zero



Key Training Facts

~5K
steps to SOTA
vs ~10K (DeepSeek)

0
training crashes
multiple runs

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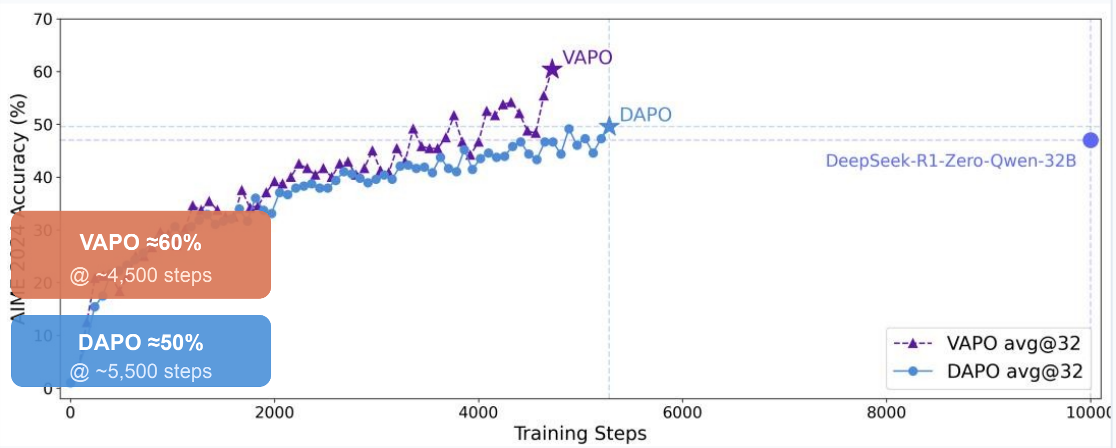
New SOTA

VAPO Reaches SOTA in ~5,000 Steps With Zero Training Crashes

VAPO achieves 60% accuracy in ~4,500 steps; DAPO plateaus at 50% by 5,500 steps; DeepSeek-R1-Zero needs ~10,000 steps for 47%.

Training Accuracy vs Steps (AIME 2024)

Source: arXiv 2504.05118, Figure 2



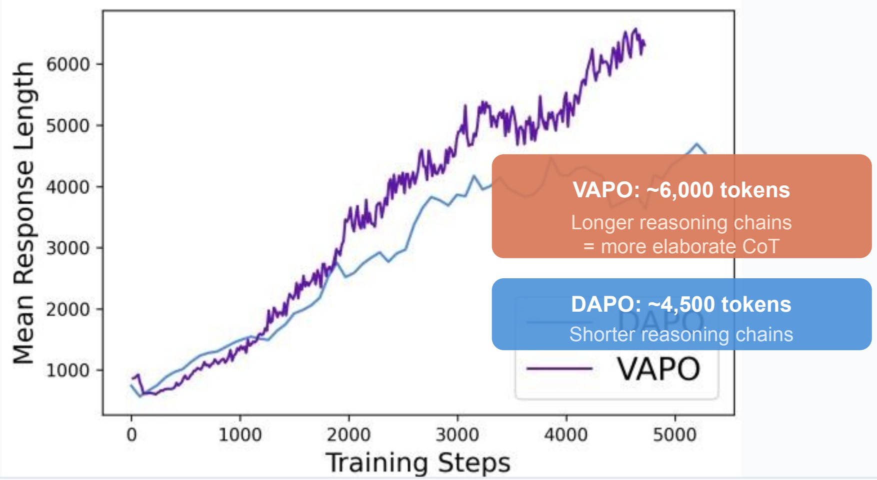
~5K
steps to SOTA
vs ~10K (DeepSeek)

2×
faster than DeepSeek
to equivalent performance

0
training crashes
multiple runs

Mean Response Length vs Training Steps

Source: arXiv 2504.05118, Figure 3



What Longer Responses Indicate

VAPO learns more elaborate reasoning chains, not just shorter correct answers. This suggests deeper exploration of the solution space.

+33% longer chains vs DAPO

Training Efficiency Summary

60%	4,500	~6K	0 crashes
VAPO final accuracy	steps to reach VAPO SOTA	avg VAPO response length	across all independent runs

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Removing Value-Pretraining collapses VAPO from 60→11. No single component can be dropped without significant performance loss.

Top 3 Critical Components

#1 Value-Pretraining −49 pts

60 → 11 without pretraining
Without it, value-model RL barely outperforms Vanilla PPO
Vanilla PPO baseline: 5 pts
Single most critical component

#2 Decoupled-GAE −27 pts

60 → 33 without Decoupled-GAE
Robust value estimation is critical
From VC-PPO, essential for stability
2nd most impactful component

#3 Length-Adaptive GAE −15 pts

60 → 45 without Length-Adaptive GAE
Novel technique, not in DAPO/GRPO
Unique VAPO innovation
3rd most impactful component

All 7 components required for VAPO = 60
Dropping any one costs at least 5 points

TAKEAWAYS

Value-Model-Based RL Is Back — With the Right Engineering

VAPO systematically addresses bias, heterogeneous lengths, and reward sparsity — achieving 60.4 on AIME 2024 with superior stability.

AIME 2024 SOTA	Training Efficiency	Value-Pretraining Impact	Training Stability
60.4	**~5K**	**+49 pts**	**Zero**
avg@32, +10 pts over DAPO	steps to SOTA (2× faster than DS)	single most critical component	crashes across all independent runs

Key Findings & Implications:

- **Value-model RL outperforms value-free methods** when bias, heterogeneous lengths, and reward sparsity are systematically addressed
- **Length-Adaptive GAE is a novel, effective technique** for heterogeneous-length reasoning tasks (removes −15 pts)
- **Value-Pretraining is non-negotiable:** without it, value-model RL barely works (60 → 11 pts)
- **VAPO sets a new open reasoning baseline** on Qwen2.5-32B without any SFT data
- **Future direction:** scale VAPO's value-model approach to larger models and diverse reasoning domains

Final Comparison (AIME 2024, Qwen2.5-32B base, no SFT):

Method	AIME 2024 Score	Training Steps	Value Model
Vanilla PPO	5	—	Yes (naive)
DeepSeek-R1-Zero-32B	47	~10,000	No
DAPO	50	~5,500	No
VAPO (Ours)	**60**	**~5,000**	**Yes (VAPO)**